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System and Method for Markerless Motion Capture

Abstract

A computing system captures markerless motion data of a user via a camera of the computing system. The computing system retargets the first plurality of points and the second plurality of points to a three-dimensional model of an avatar associated with the user, wherein the avatar is associated with an identity non-fungible token that uniquely represents the user across Web2 environments and Web3 environments, and wherein retargeting the first plurality of points and the second plurality of points animates the three-dimensional model of the avatar. The computing system renders a video local to the computing system, wherein the video comprises the markerless motion data of the user retargeted to the three-dimensional model of the avatar causing hands, face, and body of the avatar to be animated in real-time. The computing system causes a non-fungible token to be generated, the non-fungible token uniquely identifying ownership of the video.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application is a continuation of U.S. patent application Ser. No. 18/400,091, filed Dec. 29, 2023, which is a continuation of U.S. patent application Ser. No. 18/193,271, filed Mar. 30, 2023, now U.S. Pat. No. 11,860,981, issued Jan. 2, 2024, which claims priority to U.S. Provisional Application No. 63/362,179, filed Mar. 30, 2022, U.S. Provisional Application No. 63/369,734, filed Jul. 28, 2022, U.S. Provisional Application No. 63/362,860 filed Apr. 12, 2022, U.S. Provisional Application No. 63/369,735, filed Jul. 28, 2022, and U.S. Provisional Application No. 63/382,314, filed Nov. 4, 2022, all of which are hereby incorporated by reference in their entireties.

FIELD OF THE DISCLOSURE

[0002] Embodiments disclosed herein generally related to a metaverse environment.

BACKGROUND

[0003] Recently, non-fungible tokens have become a popular means for users to protect their content. For example, users are able to use a variety of platforms to tokenize their content, such that they retain ownership over that content. With the tokenized version of their content, the user can verify and transfer ownership of the content and even profit off their content by transferring the tokenized version of their content to other users.

SUMMARY

[0004] In some embodiments, a method is disclosed herein. A computing system captures markerless motion data of a user via a camera of the computing system. The capturing includes capturing a first portion of the markerless motion data of the user, wherein the first portion of the markerless motion data of the user is captured within a threshold distance of the camera, the capturing comprising identifying a first plurality of points of the user to focus on. The capturing further includes capturing a second portion of the markerless motion data of the user, wherein the second portion of the markerless motion data of the user is capture outside the threshold distance of the camera, the capturing comprising identifying a second plurality of points of the user to focus on, wherein the first plurality of points includes at least one point not included in the second plurality of points and wherein the second plurality of points includes at least one point not included in the first plurality of points. The computing system retargets the first plurality of points and the second plurality of points to a three-dimensional model of an avatar associated with the user, wherein the avatar is associated with an identity non-fungible token that uniquely represents the user across Web2 environments and Web3 environments, and wherein retargeting the first

plurality of points and the second plurality of points animates the three-dimensional model of the avatar. The computing system renders a video local to the computing system, wherein the video comprises the markerless motion data of the user retargeted to the three-dimensional model of the avatar causing hands, face, and body of the avatar to be animated in real-time. The computing system causes a non-fungible token to be generated, the non-fungible token uniquely identifying ownership of the video.

[0005] In some embodiments, a non-transitory computer readable medium is disclosed herein. The non-transitory computer readable medium includes one or more sequences of instructions, which, when executed by a processor, causes a computing system to perform operations. The operations include capturing, by a camera of the computing system, markerless motion data of a user. The capturing includes capturing a first portion of the markerless motion data of the user, wherein the first portion of the markerless motion data of the user is captured within a threshold distance of the camera, the capturing including identifying a first plurality of points of the user to focus on. The capturing further includes capturing a second portion of the markerless motion data of the user, wherein the second portion of the markerless motion data of the user is capture outside the threshold distance of the camera, the capturing comprising identifying a second plurality of points of the user to focus on, wherein the first plurality of points includes at least one point not included in the second plurality of points and wherein the second plurality of points includes at least one point not included in the first plurality of points. The operations further include retargeting, by the computing system, the first plurality of points and the second plurality of points to a three-dimensional model of an avatar associated with the user, wherein the avatar is associated with an identity non-fungible token that uniquely represents the user across Web2 environments and Web3 environments, and wherein retargeting the first plurality of points and the second plurality of points animates the three-dimensional model of the avatar. The operations further include rendering, by the computing system, a video local to the computing system, wherein the video comprises the markerless motion data of the user retargeted to the three-dimensional model of the avatar causing hands, face, and body of the avatar to be animated in real-time. The operations further include causing, by the computing system, a non-fungible token to be generated, the non-fungible token uniquely identifying ownership of the video.

[0006] In some embodiments, a system is disclosed herein. The system includes a processor and a memory. The memory has programming instructions stored thereon, which, when executed by the processor, causes the system to perform operations. The operations include capturing, by a camera, markerless motion data of a user. The capturing includes capturing a first portion of the markerless motion data of the user, wherein the first portion of the markerless motion data of the user is captured within a threshold distance of the camera, the capturing comprising identifying a first plurality of points of the user to focus on. The capturing further includes capturing a second portion of the markerless motion data of the user, wherein the second portion of the markerless motion data of the user is capture outside the threshold distance of the camera, the capturing comprising identifying a second plurality of points of the user to focus on, wherein the first plurality of points includes at least one point not included in the second plurality of points and wherein the second plurality of points includes at least one point not included in the first plurality of points. The operations further include retargeting the first plurality of points and the second plurality of points to a three-dimensional model of an avatar associated with the user, wherein the avatar is associated with an identity non-fungible token that uniquely represents the user across Web2 environments and Web3 environments, and wherein retargeting the first plurality of points and the second plurality of points animates the three-dimensional model of the avatar. The operations further include locally rendering a video, wherein the video comprises the markerless motion data of the user retargeted to the three-dimensional model of the avatar causing hands, face, and body of the avatar to be animated in real-time. The operations further include causing a non-fungible token to be generated, the non-fungible token uniquely identifying ownership of the video.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0007] So that the manner in which the above recited features of the present disclosure can be understood in detail, a more particular description of the disclosure, briefly summarized above, may be had by reference to embodiments, some of which are illustrated in the appended drawings. It is to be noted, however, that the appended drawings illustrate only typical embodiments of this disclosure and are therefore not to be considered limiting of its scope, for the disclosure may admit to other equally effective embodiments.

[0008] FIG. 1 is a block diagram illustrating a computing environment, according to one exemplary embodiment.

[0009] FIG. 2 is a block diagram illustrating server system of FIG. 1, according to example embodiments.

[0010] FIG. 3 illustrates an exemplary avatar, according to example embodiments.

[0011] FIG. 4 is a block diagram illustrating server system of FIG. 1, according to example embodiments.

[0012] FIG. 5 is a block diagram illustrating server system of FIG. 1, according to example embodiments.

[0013] FIG. 6A illustrates an exemplary avatar with accessories, according to example embodiments.

[0014] FIG. 6B illustrates an exemplary avatar with accessories, according to example embodiments.

[0015] FIG. 7 is a block diagram illustrating server system of FIG. 1, according to example embodiments.

[0016] FIG. 8 illustrates an exemplary graphical user interface, according to example embodiments.

[0017] FIG. 9A is a block diagram illustrating server system, according to example embodiments.

[0018] FIG. 9B is a block diagram illustrating server system, according to example embodiments.

[0019] FIG. 10A is a block diagram illustrating interaction between a Web2 environment and a Web3 environment, according to example embodiments.

[0020] FIG. 10B is a block diagram illustrating interaction between a Web2 environment and a Web3 environment, according to example embodiments.

[0021] FIG. 11A is a block diagram illustrating a computing environment, according to example embodiments.

[0022] FIG. 11B is a block diagram illustrating a computing environment, according to example embodiments.

[0023] FIG. 12 is a block diagram illustrating a computing environment interacting with server system 104, according to example embodiments.

[0024] FIG. 13 is a flow diagram illustrating a method of minting a digital content, according to example embodiments.

[0025] FIG. 14 illustrates an example graphical user interface (GUI), according to example embodiments.

[0026] FIG. 15 illustrates an example graphical user interface (GUI), according to example embodiments.

[0027] FIG. 16A illustrates a system bus computing system architecture, according to example embodiments.

[0028] FIG. 16B illustrates a computer system having a chipset architecture, according to example embodiments.

[0029] FIG. 17 illustrates an example diagram of a parent-child token relationship, according to

example embodiments.

[0030] FIG. **18** is flow diagram illustrating a method of generating digital content, according to example embodiments.

[0031] FIG. **19** is a block diagram illustrating a set of points, according to example embodiments.

[0032] To facilitate understanding, identical reference numerals have been used, where possible, to designate identical elements that are common to the figures. It is contemplated that elements disclosed in one embodiment may be beneficially utilized on other embodiments without specific recitation.

DETAILED DESCRIPTION

[0033] One or more techniques described herein provide a new non-fungible token platform that allows users to take ownership over digital avatars, accessories for these avatars, and digital content that the user produces. For example, one or more techniques described herein allow users to use their non-fungible token avatars in photos and videos, which they can then share to social media platforms.

[0034] The term “user” as used herein includes, for example, a person or entity that owns a computing device or wireless device; a person or entity that operates or utilizes a computing device or wireless device; or a person or entity that is otherwise associated with a computing device or wireless device. It is contemplated that the term “user” is not intended to be limiting and may include various examples beyond those described. In some embodiments, a user may be a content creator. In some embodiments, a user may be an owner of a non-fungible token of an avatar, a post, or an accessory. In some embodiments, a user can be both the owner of the non-fungible token and the content creator. In some embodiments, a user can be a person interacting with social content.

[0035] The term “metaverse” as used herein may be representative of a centralized or decentralized environment that may be accessible to end users via one or more computing devices that include, but are not limited to, mobile devices, computing devices (e.g., tablets, laptops, personal computers, etc.), wearable devices, and extended reality (XR) devices (e.g., augmented reality devices, virtual reality device). Generally, the metaverse may refer to an environment that includes both real-world components and virtual components. In some embodiments, the metaverse may refer to a single metaverse environment. In some embodiments, the metaverse may refer to multiple metaverse environments, akin to a multiverse.

[0036] The term “digital content” as used herein includes, for example, audio content (e.g., podcasts, songs, audiobooks, etc.), image content, video content, and/or any combination of one or more of audio content, image content, and video content.

[0037] FIG. **1** is a block diagram illustrating computing environment **100**, according to one embodiment. Computing environment **100** may include at least one or more user devices **102**, one or more user devices **103**, a server system **104**, a blockchain **122**, and one or more non-fungible token marketplaces **108** communicating via network **105**.

[0038] Network **105** may be of any suitable type, including individual connections via the Internet, such as cellular or Wi-Fi networks. In some embodiments, network **105** may connect terminals, services, and mobile devices using direct connections, such as radio frequency identification (RFID), near-field communication (NFC), Bluetooth™, low-energy Bluetooth™ (BLE), Wi-Fi™, ZigBee™, ambient backscatter communication (ABC) protocols, USB, WAN, or LAN. Because the information transmitted may be personal or confidential, security concerns may dictate one or more of these types of connections be encrypted or otherwise secured. In some embodiments, however, the information being transmitted may be less personal, and therefore, the network connections may be selected for convenience over security.

[0039] Network **105** may include any type of computer networking arrangement used to exchange data. For example, network **105** may be the Internet, a private data network, virtual private network using a public network and/or other suitable connection(s) that enables components in computing environment **100** to send and receive information between the components of computing

environment **100**.

[0040] User device **102** may be operated by user. User device **102** may be representative of a mobile device, a tablet, a desktop computer, virtual reality (VR) system, augmented reality (AR) system, extended reality (XR) system, or any computing system having the capabilities described herein. User device **102** may include at least application **112** and AR software **114**.

[0041] Application **112** may be representative of a social media application associated with server system **104**. In some embodiments, application **112** may be a standalone application associated with server system **104**. In some embodiments, application **112** may be representative of a web-browser configured to communicate with server system **104**. In some embodiments, user device **102** may communicate over network **105** to request a webpage, for example, from web client application server **118** of server system **104**. For example, user device **102** may be configured to execute application **112** to enter a social media website existing on the metaverse and hosted by web client application server **118**. The content that is displayed to user device **102** may be transmitted from web client application server **118** to user device **102**, and subsequently processed by application **112** for display through a graphical user interface (GUI) of user device **102**.

[0042] AR software **114** may be representative of a native or third party software that allows users to create augmented reality images or videos. In some embodiments, AR software **114** may be representative of one or more of Google's markerless motion tracking software; ARKit®, which is a software development kit (SDK) commercially available from Apple®, MediaPipe, which is a machine learning base that can be customized and is commercially available from Google®; ARCore, which is a software development kit developed by Google®, or the like, or any combination thereof, that allows users to, in part, generate augmented reality videos and/or images. A user of user device **102** may use AR software **114** to generate a video or image of themselves and replace themselves in the video or image with their own digital avatar or insert their own digital avatar into the video or image. These videos and/or images may be posted to a user's social media account with metaverse platform **120** via application **112**.

[0043] In some embodiments, rather than having AR software **114** be separate from application **112**, application **112** may include AR software **114**. In other words, a user may utilize AR software of application **112** to generate videos or images with their own digital avatar.

[0044] Generally, AR software **114** may provide for live tracking of the body, the face, and hands of a user or users. Compared to existing AR software solutions, AR software **114** is the first markerless motion capture software that allows for live tracking of the body, the face, and the hands simultaneously. In operation, AR software **114** may be configured to extract over a plurality of reference points or landmarks (e.g., at least 30) from an individual captured within a field of view of a camera **119** of user device **102**. Using the plurality of reference points, AR software **114** may access one or more third party APIs or SDKs such that a skeleton structure of the user can be generated. Once the skeleton structure of the user is generated, AR software **114** may locally retarget, in real-time or near real-time, data in the skeleton structure to the avatar to animate the avatar. In this manner, AR software **114** may allow a user to locally render avatars on user device **102**.

[0045] In conventional systems, AR software typically has difficulty when toggling between distance shots and close-range shots. To account for this, AR software **114** may include an artificial intelligence module **115**. Artificial intelligence module **115** may be configured such that, when user device **102** is in close range of the user (e.g., when the user's face fills up a threshold percentage of the screen), artificial intelligence module **115** may be configured to lock the chest of the user to the bottom of the screen. Once the user's body is locked in, AR software **114** may then prioritize focusing on the face and hands of the user, instead of searching for the rest of the user's body. In some embodiments, rather than locking the user's body, there may be a smooth transition when the user moves from close range to far range or vice versa.

[0046] Artificial intelligence module **115** may be trained to determine how to focus a camera **119** of

user device **102**, depending on if the user is triggering a selfie mode or a full body mode. For example, in full body mode, artificial intelligence module **115** may be trained to identify a plurality of anchor points on the user's body. AR software **114** may focus less on face anchor points when the user's full body is in frame. In such example, user may be able to open and close their mouth and eyes, smile, or frown; but user may not be able to make as many facial expressions as they otherwise would in selfie mode. However, in full body mode, because artificial intelligence module **115** may instruct AR software **114** to focus less on processing power toward the user's face, AR software **114** may be able to capture the hands and rest of the user's body more accurately. In selfie mode, rather than change the plurality of anchor points, artificial intelligence module **115** may be trained to ignore a subset of anchor points, depending on the distance of the user from camera **119**. For example, AR software **114** may ignore all leg anchor points when in selfie mode and artificial intelligence module **115** may instruct AR software **114** to focus less on the lower body and more on the user's face, so AR software **114** may be able to capture facial gestures more accurately.

[0047] Further, in most conventional technology that is publicly available is limited to only tracking the user's face or generally tracking the user's body without more precise finger tracking and face tracking. AR software **114** improves upon the current state of the art by providing the ability to merge body tracking from head to toe, precise finger tracking, and more precise face tracking. For example, AR software **114** may provide the ability for a user to make two peace signs next to their face, and have a model overlaid to reflect that on their phone screen using their phone camera **119** or any other sensors available to user device **102** (e.g., LIDAR sensor). AR software **114** may thus be able to track all of the user's hand, face, and body movements in real time.

[0048] To facilitate such approach, AR software **114** may manage several different PointPools, with each pool representing a different body part. In some embodiments, there are four main PointPools being used: Body (head, arms, legs, feet), LHand (left hand), RHand (right hand), Face (facial landmarks).

[0049] AR software **114** may fill these point pools through, for example, a multithreaded process running in the background. Each RGB frame may be captured from a designated streaming camera and stored in a texture object using a GPU command. AR software **114** may provide these texture objects to, for example, one or more computer vision SDKs and/or APIs to process. From these textures, AR software **114** may generate landmarks, which are structures of unfiltered vectors representing what AR software **114** may think the body points of a person on the camera should be in 3D space. AR software **114** may take those landmark points and translate them to a compatible vector for processing (e.g., Unity Vector3s).

[0050] In some embodiments, each point or point pool may be designated its own processing pipeline through which the points flow through. In some embodiments, the points in their raw form are too unreliable to be sent straight into the inverse kinematics system. Accordingly, the pipelines may be utilized to make corrections to the points. In some embodiments, the pipelines may include low pass filters, such as Kalman filters and box filters in order to filter out the noise and inaccuracies from the machine learning. In some embodiments, the pipelines may also include multiple correction objects specific to certain body PointPools to represent the points as accurately in 3D space as possible. The pipelines may interact with each other to correct multiple different parts of the body using other parts as points of reference.

[0051] Once the points have been corrected and adjusted by the pipelines, AR software **114** may compile them into a single object containing all the points and may use an inverse kinematics (IK) system to calculate the proper model transformations. The IK system may include two sets of classes, one of which polls this object per frame to get the most recent body positions, and another that sets the positions and rotations of the model. Two separate calculations may be used to set the body transforms based on the distance of the source person from camera **119**. When the person on camera **119** is far, AR software **114** may use a Quaternion based IK system that rotates the body segments so that the body's forward vector matches the machine learning's corrected points'

forward vector. In some embodiments, server system **104** may utilize machine learning to train AR software **114** to anticipate user rotation and placement of the body to accurately predict how the user may move when recording a video and improve filters.

[0052] In some embodiments, such as when the person is close or the model being retargeted has unique proportions, AR software **114** may use additional or alternate filter systems, such as forward and backward reaching inverse kinematics (FABRIK) instead of the default IK system in place.

[0053] In some embodiments, AR software **114** may use a combination of the IK system with ARKit or ARCore, trading in performance for more stability and accuracy. In some embodiments, separate pipelines can be made by filtering in point data from those two sources if the device can handle it.

[0054] In some embodiments, AR software **114** may utilize forward kinematics or other methods.

[0055] In some embodiments, AR software **114** may be configured to capture and retarget data to animate a user's asset (e.g., avatar token). Such data from the user's movements can be retargeted. For example, such movement information may be retargeted to gamify social media. For example, on a social media platform, such as TikTok, AR software **114** may be used to capture user motion data from movements in dancing trend videos or posts. The movement data can be retargeted to determine accuracy in movement or other metrics based on the movement captured and the game being played. In some embodiments, AR software **114** may be configured to issue scores based on the most accurate "best" dance.

[0056] In some embodiments, AR software **114** may be configured to animate multiple objects in addition to multiple people. For example, AR software **114** may be configured to recognize a soccer ball and separately track the soccer ball. In some embodiments, rather than AR software **114** being utilized to recognize objects, an artificial intelligence engine or computer vision engine may be trained to recognize objects and separately track those objects, with or without sensors or the object. Such information may be used to provide feedback regarding the movement of those objects.

[0057] In some embodiments, AR software **114** may be used to animate objects in the real world to drive users on a trail, hunt, or to a specific retail location.

[0058] In some embodiments, the process of rendering a user's digital avatar on a video or image generated by user device **102** using AR software **114** may involve a hybrid approach that combines real-time local rendering with higher quality remote rendering or delayed local rendering. For example, a user can capture a video or image using client device **102**. Client device **102** may perform real-time local rendering of the image or video while the user is capturing the video or image. For example, AR software **114** may perform a real-time local rendering of the user's digital avatar. This rendering may utilize one or more techniques discussed above, such as three-dimensional modeling, image-based rendering, or depth-based rendering. The purpose of this real-time rendering is to provide a quick preview of the content, allowing the user to see how their digital avatar will appear in the final output. In some cases, the preview of the content may be sufficient, and the user may save and utilize the content rather than choosing to wait for the final, delayed output.

[0059] In some embodiments, in addition to the real-time local rendering, AR software **114** may either send the user's video or image to a remote server (e.g., server system **104**) or perform a delayed local rendering on the user device **102**. The remote server or the delayed local rendering process may be responsible for producing a higher quality version of the user's digital avatar and movements. When the remote server rendering option is used, server system **104** may receive the user's video or image, process it using more advanced rendering techniques or computational resources, and generate a higher quality rendering of the user's digital avatar and movements. This process may involve leveraging powerful GPUs, neural networks, or other hardware and software resources to enhance the quality of the avatar rendering. Once the server system **104** has completed rendering the high-quality digital avatar, or the delayed local rendering process is complete, the

final output is either sent back to user device **102** or saved locally on user device **102**. The user can now view, share, or interact with the higher quality rendering of their digital avatar overlaid on the original video or image, or in another environment of their choosing. By employing a hybrid approach, AR software **114** may provide users with a real-time preview of their digital avatar in the captured content, while also ensuring a higher quality output is available later. This approach takes advantage of the strengths of both local and remote rendering or delayed local rendering, offering a balance between real-time interaction and high-quality results. In some embodiments, the raw point pool data may be saved separately and utilize this hybrid approach to smooth any inconsistencies, tracking errors, and missing data points for a higher fidelity render. This approach requires less data to be saved or cached and provides expedited upload and rendering speeds.

[0060] FIG. **19** is a block diagram illustrating a set of points, according to example embodiments. As shown, FIG. **19** includes a first set of points **1902** and a second set of points **1904** on a user's face. First set of points **1902** may represent a set of points prior to the pipeline adjustments. Second set of points **1904** may represent a set of points after the pipeline adjustments. As seen, first set of points **1902** of the user's face appears flat and may lack depth. First set of points **1902** may pass through the one or more filters, as discussed above, to generate second set of points **1904**. As shown, second set of points **1904** may better illustrate depth and rotation of the user's face. As those skilled in the art understand, such process can be applied to other parts of the user's body.

[0061] As shown, in some embodiments, each user device **102** may be in communication with a respective cryptographic wallet **116**. Cryptographic wallet **116** may be configured to store the user's fungible tokens (e.g., cryptocurrency) and non-fungible tokens in an encrypted manner. In some embodiments, cryptographic wallet **116** may be representative of a hot wallet (e.g., Coinbase, Metamask, etc.), which is a cryptographic wallet **116** connected to network **105**. In some embodiments, cryptographic wallet **116** may be representative of a cold wallet (e.g., Ledger Nano X, Trezor Model T, and the like). In some embodiments, cryptographic wallet **116** may be incorporated within application **112**. For example, rather than a user storing their fungible tokens and/or non-fungible tokens via a third party wallet, the user may leverage a cryptographic wallet associated with server system **104** via application **112**.

[0062] User device **103** may be operated by user. User device **103** may be representative of a mobile device, a tablet, a desktop computer, virtual reality (VR) system, augmented reality (AR) system, extended reality (XR) system, or any computing system having the capabilities described herein. User device **103** may include at least application **132**.

[0063] Application **132** may be representative of a social media application associated with server system **104**. In some embodiments, application **132** may be a standalone application associated with server system **104**. In some embodiments, application **132** may be representative of a web-browser configured to communicate with server system **104**. In some embodiments, user device **103** may communicate over network **105** to request a webpage, for example, from web client application server **118** of server system **104**. For example, user device **103** may be configured to execute application **132** to enter a social media website existing on the metaverse and hosted by web client application server **118**. The content that is displayed to user device **103** may be transmitted from web client application server **118** to user device **103**, and subsequently processed by application **132** for display through a graphical user interface (GUI) of user device **103**.

[0064] Server system **104** may be configured to host a metaverse application accessible to user devices **102** and user devices **103**. For example, server system **104** may include web client application server **118** and metaverse platform **120**. Metaverse platform **120** may be configured to host digital content generated by a user and facilitate digital interaction with the digital content within a metaverse environment. For example, metaverse platform **120** may be configured to host a simulated virtual environment in which users can post images or videos of their avatar within the simulated virtual environment. For example, metaverse platform **120** may allow users to take ownership over digital avatars, accessories for these avatars, and social content that users produce

whether they use these avatars or not.

[0065] In some embodiments, a user may access metaverse environment **124** using one or more third party authentication techniques. For example, a user may access metaverse environment **124** by using “Login with Ethereum virtual machine (EVM) compatible wallet” functionality, thus allowing the user to be verified using their Ethereum accounts.

[0066] In some embodiments, a user of metaverse platform **120** may include a unique avatar. The user's unique avatar may be minted as a non-fungible token such that no two users have the same exact avatar. In some embodiments, each avatar may include certain traits that may be treated as default traits but could ultimately be replaced. For example, each avatar may include default clothing or default hair, which ultimately can be replaced by the owner. Using AR software **114**, a user can create digital content to post to metaverse platform **120**. The content may be representative of a video data, audio data, and/or video data. In some embodiments, the digital content may include an image or video of the user, in which the user's non-fungible token avatar is used in the video or image.

[0067] When a user creates a post to metaverse platform **120**, in some embodiments, the digital content may be immediately represented as a non-fungible token through a process referred to as “post-to-mint.” In some embodiments, initially, the non-fungible token is not minted onto blockchain **122** until a transaction takes place. A transaction may refer to a buying or selling of the non-fungible token corresponding to the user's post. Such process may provide users with a gasless mint process for the users, while also instantaneously allowing users to have a marketable product in the form of a non-fungible token.

[0068] The digital content may be displayed on the user's profile, the profile of the owner of the content, a licensee of the content, a tagged user in the content, or the like. In some embodiments, other users and/or community members can interact with the post on the user's profile. In some embodiments, interactions may come in the form of engagements or offers.

[0069] In some embodiments, metaverse platform **120** may support a “use-to-earn” model. For example, metaverse platform **120** may facilitate organic engagements (i.e., user interactions) to create activity and drive adoption of the platform, as well as reward or otherwise provide value to creators and owners of content. To do so, each post may be associated with a score. The score may be representative of a cumulative score of how many likes, shares, and other interactions a post receives. In some embodiments, certain activities will be rewarded with a higher score, ad points, and/or fungible tokens unique to metaverse platform **120** (e.g., “utility token”). In this manner, engagements such as likes, shares, views/plays, and comments can create a financial incentive for the end-user.

[0070] In some embodiments, the utility token may take the form of an ERC20 token broadcast or written to blockchain **122**. For example, the utility token may follow the Standard Openzeppelin Mintable ERC20 standard. The utility tokens may initially be held by an entity associated with server system **104**. For example, the ERC20 tokens may be deployed from and owned by the entity using a Gnosis Multisig wallet, thus allowing administrators to retain control over the tokens until transferred to a user.

[0071] Accordingly, unlike conventional metaverse or social media environments, metaverse platform **120** may function as a social media platform in which a user's avatar and posts can be represented by non-fungible tokens. For example, upon posting an image or video to metaverse platform **120**, metaverse platform **120** may mint a non-fungible token to tokenize the image or video. In this manner, metaverse platform **120** can provide users with the ability to protect and market their digital content through non-fungible tokens.

[0072] Non-fungible token marketplaces **108** may be representative of one or more online marketplaces in which a user can upload or link their metaverse digital content. Via non-fungible token marketplaces **108** a user can buy, sell, or showcase their tokenized items of digital content. In some embodiments, non-fungible token marketplaces **108** may include centralized marketplaces

(e.g., OpenSea) and/or decentralized marketplaces (e.g., LOOKSRARE). In some embodiments, any post within metaverse platform **120** that is also linked to a non-fungible token marketplace **108** may include a uniform resource locator or link that links back to the original post in metaverse platform **120**.

[0073] Networked computers **106** may be configured to host blockchain **122**. In some embodiments, blockchain **122** may be a public blockchain. In some embodiments, blockchain **122** may be a private blockchain, such as, for example, a private blockchain associated with server system **104**. Generally, blockchain **122** may be representative of any blockchain configured to support non-fungible tokens. For example, blockchain **122** may be a public or private blockchain based on the Ethereum platform.

[0074] FIG. **2** is a block diagram **200** illustrating server system **104**, according to example embodiments. As shown, server system **104** includes repository **202** and one or more computer processors **204**.

[0075] Repository **202** may be representative of any type of storage unit and/or device (e.g., a file system, database, collection of tables, or any other storage mechanism) for storing data. Further, repository **202** may include multiple different storage units and/or devices. The multiple different storage units and/or devices may or may not be of the same type or located at the same physical site. As shown, repository **202** includes at least metaverse platform **120**.

[0076] Metaverse platform **120** may include an intake module **206**, a tokenization module **210**, and a broadcast module **214**. While metaverse platform **120** may include more modules, the present discussion focuses on an initial avatar generation process. Each of intake module **206**, tokenization module **210**, and broadcast module **214** may be comprised of one or more software modules. The one or more software modules are collections of code or instructions stored on a media (e.g., memory of server system **104**) that represent a series of machine instructions (e.g., program code) that implements one or more algorithmic steps. Such machine instructions may be the actual computer code the processor of server system **104** interprets to implement the instructions or, alternatively, may be a higher level of coding of the instructions that are interpreted to obtain the actual computer code. The one or more software modules may also include one or more hardware components. One or more aspects of an example algorithm may be performed by the hardware components (e.g., circuitry) itself, rather than as a result of the instructions.

[0077] As shown, a user device **102** may upload or provide metaverse platform **120** with an avatar image indicative of an avatar of the user. In some embodiments, the avatar image may be used to uniquely identify the user within the metaverse hosted by metaverse platform **120**.

[0078] Intake module **206** may be configured to receive the avatar image and metadata information from user device **102**. Upon receiving the avatar image of the user's avatar, intake module **206** may store the image at a network accessible location. For example, intake module **206** may store the avatar image storage location **208**. Although storage location **208** is shown internal to metaverse platform **120**, those skilled in the art understand that storage location **208** may represent an external file hosting server or service. In some embodiments, storage location **208** may be controlled by an entity associated with server system **104**. In some embodiments, storage location **208** may be an external service, such as, but not limited to, InterPlanetary File System (IPFS). Upon storing the image in storage location **208**, intake module **206** may generate or receive a unique uniform resource identifier (URI) indicating the storage location of the avatar image. Intake module **20** may provide the URI of the image to the user of user device **102**. The user may use the URI of the avatar image to generate a metadata file corresponding to the avatar image.

[0079] In some embodiments, the user may upload a metadata file corresponding to the avatar image. The metadata file may define the traits for the avatar image. Exemplary traits may include, but are not limited to, base traits, a tech pack, regional traits (e.g., tech lines, clip accents, main region, etc.), eyes, teeth, headwear, ears, and the like. Through the combination of traits, there may be more than a quadrillion combinations of traits. Accordingly, there may be more than a

quadrillion unique avatars within the metaverse environment. In some embodiments, the metadata file may be representative of, for example, a JSON file, to which the eventual non-fungible token will be linked. The metadata file may include, for example, the URI that identifies storage location **208** of the avatar image. Intake module **206** may further store the metadata file corresponding to the avatar image in storage location **208**.

[0080] Tokenization module **210** may be configured to tokenize the avatar image provided by the user. For example, tokenization module **210** may be configured to generate a smart contract based on the avatar image. In some embodiments, the smart contract may be representative of an ERC721 token. The smart contract may include, for example, a URI corresponding to token **212**, the URI that identifies a storage location of the avatar image in storage location **208**, and the URI that identifies a storage location of the metadata file corresponding to the avatar image.

[0081] Broadcast module **214** may be configured to broadcast or write token **212** to blockchain **122**. In some embodiments, broadcast module **214** may directly write token **212** to blockchain **122**. For example, in some embodiments, broadcast module **214** may deploy token **212** on Ethereum Mainnet. Once deployed, token **212** may be bridged off Ethereum Mainnet to cheaper layer-2 blockchains (L2) like Polygon. Such bridging may make it easier for users to interact directly from an L2.

[0082] As those skilled in the art understand, to mint the user's avatar, a user may have to pay the gas fee associated with the minting process. The gas fee may provide miners with an incentive to process and validate transactions on blockchain **122**.

[0083] Once the avatar image is minted, token **212** may be provided back to user device **102** for storage in the user's wallet **116**. In some embodiments, token **212** may be temporarily held or controlled by an entity associated with server system **104**. For example, token **212** may initially be deployed from and owned by the entity, thus allowing a development team to retain control until token **212** is transferred from a wallet associated with server system **104** (e.g., Gnosis Multisig) to a wallet **116** of the user.

[0084] Although the above discussion focuses on an avatar being minted as a token **212**, those skilled in the art understand that such avatars need not be minted as a token.

[0085] In some embodiments, rather than allowing an end user (e.g., user) to create their own digital avatars, the above process may be instead performed by an administrator of server system **104**. For example, an administrator may generate an initial plurality of avatars by randomly selecting combinations of traits. Such initial plurality of avatars may be made available to end users through an auction process. For example, server system **104** may use a traditional Dutch auction process, in which avatars may be rewarded to end users. In such embodiments, server system **104** may initially generate a convertible token using the ERC1155 standard. Once an end user is rewarded with an avatar through the auction process, the end user may be instructed to “burn” the ERC1155 token, which would then exchange the convertible token from the ERC1155 token to the ERC721 token.

[0086] In some embodiments, metaverse platform **120** may support replacement of users with their avatar equivalent in real-time during the content creation process or after the original posting of the generated content. For example, if Bob is a user of metaverse platform **120** and has an associated avatar, metaverse platform **120** may support facial recognition processes that identifies Bob in social media content posted to metaverse platform **120** and replaces the photo of Bob with their associated avatar.

[0087] FIG. **3** illustrates an exemplary avatar **300**, according to example embodiments. As shown, avatar **300** may be representative of an avatar that is uniquely associated with a user or user account. Avatar **300** may be representative of a base character without any accessories. As shown, the appearance of avatar **300** may be dependent on the metadata defined by the user. For example, a user may define, in the metadata file associated with avatar **300**, certain metadata categories that influence the appearance of avatar **300**.

[0088] Avatar **300** may include a plurality of customizable regions. For example, a user may be able to customize one or more of: the base body, base face, base skin tone, a plurality of regions where various textures and colors can be applied, eyes, ears, and the like. Each region of the plurality of regions may include different textures and/or colors. Each region may have an associated metadata column, in which that region's textures and colors can be defined. In some embodiments, the metadata file associated with avatar **300** may further include a special illumination field. The special illumination field may be used for rare traits. For example, the special illumination field can allow for rare traits, such as tech lines lighting up. As those skilled in the art understand, such avatar **300** may be used outside of a metaverse environment, such as in different metaverse environments or non-metaverse environments. For example, a user can use their avatar **300** on other platforms, such as, but not limited to, a video game environment (e.g., a custom character in a game), a streaming service, and the like. For example, a user may use their avatar **300** as a drop-in replacement for themselves in a video conference, stream, live render, etc. In this manner, any video generated through the video conference, stream, or live render, using avatar **300**, may itself be minted as a token.

[0089] In some embodiments, there may be multiple displays of avatar **300**. For example, a user may record and stream a live feed of themselves, while they are being converted to their avatar **300** in real-time or near real-time. For example, a user may be playing on stage, while being projected as an MTskin in all surrounding displays. Simultaneously, the output of the MTskin may be streamed to be the performer on stage in a metaverse environment. Thus, all surrounding displays in the metaverse would show a feed of the physical performer.

[0090] In operation, when a user accesses their avatar (e.g., avatar **300**) via application **112**, application **112** may extract metadata information for the avatar from cryptographic wallet **116**. As such, application **112** may render the user's avatar locally on user device **102** by reconstruction the user's avatar using the base avatar character and the defined traits for the avatar, as stored in cryptographic wallet **116**.

[0091] FIG. **4** is a block diagram **400** illustrating server system **104**, according to example embodiments. As shown, block diagram **400** may illustrate components of server system **104** involved in the minting of accessories for the user's avatar.

[0092] As shown, a user device **102** may upload or provide metaverse platform **120** with an accessory image indicative of an accessory for user with the user's avatar. Exemplary accessories may be, for example, hats, shirts, watches, wallets, bags, swords, basketballs, etc. The accessories may be used to customize or change the appearance of the user's avatar in digital content.

[0093] Intake module **206** may be configured to receive the accessory image and metadata information from user device **102**. Upon receiving the accessory image, intake module **206** may store the image at a network accessible location. For example, intake module **206** may store the accessory image at storage location **208**. Upon storing the image in storage location **208**, intake module **206** may generate or receive a unique uniform resource identifier (URL) indicating the storage location of the accessory image. Intake module **206** may provide the URL of the accessory image to the user of user device **102**. The user may use the URL of the accessory image to generate a metadata file corresponding to the accessory image.

[0094] In some embodiments, the user may upload a metadata file corresponding to the accessory image. The metadata file may define the traits for the accessory image. In some embodiments, the metadata file may be representative of a JSON file to which the eventual non-fungible token will be linked. The metadata file may include, for example, the URL that identifies storage location **208** of the accessory image. Intake module **206** may further store the metadata file corresponding to the accessory image in storage location **208**.

[0095] Tokenization module **210** may be configured to tokenize the accessory image provided by the user. For example, tokenization module **210** may be configured to generate a smart contract based on the accessory image. In some embodiments, the smart contract may be representative of

an ERC1155 token. Tokenization module **210** may utilize an ERC1155 token in order to make the accessories tradable and handle the semi-fungibility of accessories. Separate ERC1155 tokens for each accessory may allow for new accessories to be released to match the needs of that specific accessory's launch but can still interact with the existing posts. The smart contract may include, for example, a URI corresponding to token **412**, the URL that identifies a storage location of the accessory image in storage location **208**, and the URL that identifies a storage location of the metadata file corresponding to the accessory image.

[0096] Broadcast module **214** may be configured to broadcast or write token **412** to blockchain **122**. In some embodiments, broadcast module **214** may directly write token **412** to blockchain **122**. For example, in some embodiments, broadcast module **214** may deploy token **412** on Ethereum Mainnet. Once deployed, token **412** may be bridged off Ethereum Mainnet to cheaper L2s.

[0097] As those skilled in the art understand, to mint the accessory, a user may have to pay the gas fee associated with the minting process. The gas fee may provide miners with an incentive to process and validate transactions on blockchain **122**.

[0098] Once the accessory image is minted, token **412** may be provided back to user device **102** for storage in the user's wallet **116**. In some embodiments, token **412** may be temporarily held or controlled by an entity associated with server system **104**. For example, token **412** may initially be deployed from and owned by the entity, thus allowing a development team to retain control until token **412** is transferred from a wallet associated with server system **104** (e.g., Gnosis Multisig) to a wallet **116** of the user.

[0099] FIG. 5 is a block diagram **500** illustrating server system **104**, according to example embodiments. As shown, block diagram **500** may illustrate components of server system **104** involved in the minting of digital content.

[0100] As shown, a user device **102** may upload or provide metaverse platform **120** with digital content. In some embodiments, the digital content may be representative of image content and/or video content generated by the user using user device **102**. For example, a user may generate digital content using AR software **114**. By using AR software **114**, the user can insert the user's tokenized avatar into audio and/or video content.

[0101] Intake module **206** may be configured to receive the digital content and metadata information corresponding to the digital content from user device **102**. Upon receiving the digital content, intake module **206** may store the digital content at a network accessible location. For example, intake module **206** may store the content in storage location **208**. Upon storing the digital content in storage location **208**, intake module **206** may generate or receive a URL indicating the storage location of the digital content. Intake module **206** may provide the URL of the digital content to the user of user device **102**. The user may use the URL of the digital content to generate a metadata file corresponding to the digital content.

[0102] In some embodiments, the user may upload a metadata file corresponding to the digital content to intake module **206**. The metadata file may define the traits for the digital content. Exemplary traits may include, but are not limited to, base traits, a tech pack, regional traits (e.g., tech lines, clip accents, main region, etc.), eyes, teeth, headwear, ears, and the like. In some embodiments, the metadata file may be representative of a JSON file to which the eventual non-fungible token will be linked. The metadata file may include, for example, the URI that identifies storage location **208** of the digital content. Intake module **206** may further store the metadata file corresponding to the digital content in storage location **208**.

[0103] As indicated above, the digital content produced by the user includes the user's avatar token. In order to link the user's avatar token with the digital content, the metadata file may further include the URI corresponding to the tokenized avatar.

[0104] In some embodiments, the digital content produced by the user may further include an accessory token made available from server system **104** and/or generated by the user. In order to link the accessory token with the digital content, the metadata file may further include the URI

corresponding to the tokenized accessory.

[0105] Tokenization module **210** may be configured to tokenize the digital content provided by the user (i.e., token **512**). For example, tokenization module **210** may be configured to generate a smart contract based on the digital content and the avatar image. In some embodiments, the smart contract may further be based on any indicated accessories. In some embodiments, the smart contract may be representative of an ERC721 token. In some embodiments, the smart contract may be representative of an ERC1155 token. The smart contract may include, for example, a URI corresponding to token **512**, the URL that identifies a storage location of the digital content in storage location **208**, and the URL that identifies a storage location of the metadata file corresponding to the digital content. In some embodiments, the smart contract may further include a URI corresponding to token **212**, the URL that identifies a storage location of the avatar image in storage location **208**, and the URL that identifies a storage location of the metadata file corresponding to the avatar image. In some embodiments, the smart contract may further include a URI corresponding to token **412**, the URL that identifies a storage location of the accessory image in storage location **208**, and the URL that identifies a storage location of the metadata file corresponding to the accessory image.

[0106] As those skilled in the art recognize, when a user mints a non-fungible token, the user typically has to pay a gas fee for the minting process to incentive miners to process and validate transaction on blockchain **122**. The gas fee may disincentivize users from generating posts within the metaverse environment. Accordingly, to incentivize users to generate content within the metaverse environment, content posts may undergo a “gasless mint.” In a gasless mint, tokenization module **210** may generate the ERC721 or ERC 1155 token corresponding to the digital content; however, broadcast module **214** may not broadcast or write token **512** to blockchain **122** until another user interacts with the digital content. For example, if another user buys the digital content, such action may trigger the writing of token **512** to blockchain **122**. In other words, token **512** may not be broadcast or written to blockchain **122** until another user buys token **512**. In this manner, the burden of paying the gas fee may be transferred from the user to the buyer of the digital content.

[0107] Accordingly, once token **512** is generated, token **512** may be stored in storage location **208**. Broadcast module **214** may be configured to broadcast or post the digital content to the user's account within the metaverse environment and/or one or more external non-fungible token marketplaces **108** (e.g., OpenSea).

[0108] As those skilled in the art understand, in some embodiments, the gasless minting process can be replaced with a direct minting process, whereby the digital content is instantly minted to blockchain **122** and the associated gas fee is paid by the user.

[0109] In some embodiments, token **512** may be owned by multiple users. For example, shares of token **512** may be fractionalized, such that token **512** may have a plurality of owners. For example, user **1** owns a 30% share, user **2** owns a 50% share, and user **3** owns a 20% share. In this manner, any revenue garnered from token **512** (e.g., through interactions or engagements) may be distributed among the shareholders of token **512**. In this manner, tokenized social media content may be fractionalized, such that ownership of the tokenized social content may be shared among multiple owners.

[0110] In some embodiments, once a post is generated, a user may tag that post with a location. For example, a user's post may be taken at City Hall and the user may tag City Hall as the location for the post. In this manner, when another user accesses metaverse environment **124** via user device **103**, the user may view the post within metaverse environment **124** when approaching City Hall. For example, when digital content is minted and tagged with a geolocation, other users may access an interactive map that details the location of the digital content within the metaverse environment. If, for example, a user accesses metaverse environment using an extended reality device, the user may be able to see the post in the physical world by using the extended reality software.

Accordingly, when the user is near City Hall in the physical world, the post may be visible to the user via the extended reality software.

[0111] In some embodiments, once a post is generated, a user viewing that post may choose to view it in their physical space. For example, a user may generate a post of their avatar dancing in a park, the user viewing the generated post can select to view the avatar, using the extended reality software, dancing on the coffee table of their living room.

[0112] In some embodiments, any of tokens **212**, **412**, or **512** may be a “parent” token or a “child” token. For example, a parent token may have, stemming off of it, one or more child tokens.

[0113] Using a more specific example, an ownership token (e.g., a parent token) can be stored a more secure form of storage (e.g., cold storage), while the child token associated with the parent token can be connected, for example, to a social media platform (e.g., Twitter) as a profile picture, used for utilizing or playing as your avatar in a video game, or lent as part of a licensing deal to be used in social media, games, etc. Such processes can be performed without ever having to send the parent token. In some embodiments, the child token can be tied to the parent token on the same blockchain or even another blockchain. In this manner, there are far fewer risks associated going cross platform and cross chain since the child can be recalled. In some embodiments, the tokenized data can further be anonymized by utilizing, for example, a zero knowledge proof (or zero knowledge rollup) to create an encrypted layer between the parent token and child token for privacy purposes.

[0114] In some embodiments, a token may be a “child” token. For example, the token may be tethered to another parent token. In some embodiments, a child token may not be transferrable on its own without the parent token. In some embodiments, a child token may be transferrable on its own without the parent token. If a user owns a parent token, the user can claim the child token regardless of whether another person holds it. Such functionality may provide users with another set of security. For example, the user's parent token can be stored in a more secure vault (e.g., cold storage) while a child token tethered to the parent token remains in a hot storage wallet.

[0115] In some embodiments, such parent/child relationship may be utilized to create personalized advertisements that may directed to the consumer but prevent that personal information from reaching the advertiser. Such functionality may be a practical way of allowing users to opt into seeing certain advertisements, but with a layer of anonymization, security, and further customization of targeted advertisements the user is being served, such as by wallet address.

[0116] In some embodiments, a user may be able to lease or loan out their avatars or accessories. For example, as discussed above, an avatar token **212** or accessory token **412** may be representative of a parent token that includes a plurality of child tokens. The user (i.e., lessor) may facilitate a smart contract with another user (e.g., a lessee) that allows the lessee to temporarily use the user's avatar or accessory in metaverse environment **124** or outside of metaverse environment **124** (e.g., video game, Web2 environment, etc.).

[0117] FIG. **17** illustrates an example diagram of a parent-child token relationship, according to example embodiments. As shown, there is a parent token **1702**. Parent token **1702** may include n child tokens associated therewith (token **1704.sub.1**, **1704.sub.2**, . . . , **1704.sub.n**) (generally child token **1704**). Each child token **1704** may be leased out or loaned out to another user. For example, each child token **1704** may have corresponding metadata **1706** associated therewith. Metadata **1706** may define the terms of the lease agreement between lessor and lessee.

[0118] In some embodiments, rather than utilize a parent/child token relationship, a user may be able to lease or loan out their avatars using another user as a temporary custodian of avatar token **212** or accessory token **412**. For example, if a parent token is licensed to be used in marketing material, the smart contract could restrict the movement of either token until the term or certain conditions are met.

[0119] In some embodiments, metaverse platform **120** may be configured to take any metadata that may be integrated in a database associated with metaverse platform **120** and may reproduce that

asset or metadata inside metaverse platform **120**. Through the rendering of metadata, users may have the ability to choose which assets may be represented in the media output. Such functionality may provide users with the ability to choose any asset they own or have access to and, in real-time, have that asset render inside a three-dimensional environment, such as a metaverse environment. Once the asset is rendered, metaverse platform **120** may pull skeletal data of the user in a frame and animate the chosen avatar in real-time or near real-time.

[0120] FIG. **6A** illustrates an exemplary avatar **600** with accessory **602**, accessory **604**, and accessory **606**, according to example embodiments. As shown avatar **600** may be based on base character avatar **300** discussed above in conjunction with FIG. **3**.

[0121] Avatar **600** may include a plurality of accessories **602-606** associated therewith. As shown, avatar **600** may include accessory **602** corresponding to a glasses accessory, accessory **604** corresponding to a pants accessory, and accessory **606** correspond to a boots accessory. Each of accessory **602-606** may be represented as its own respective accessory token. Such accessories **602-606** may be rendered on the base avatar. For example, such accessories **602-606** may be rendered on the base avatar, locally, on user device **102**.

[0122] FIG. **6B** illustrates an exemplary avatar **650** with accessory **652**, accessory **654**, and accessory **656**, according to example embodiments. As shown avatar **650** may be based on base character avatar **300** discussed above in conjunction with FIG. **3**.

[0123] Avatar **650** may include a plurality of accessories **652-656** associated therewith. As shown, avatar **650** may include accessory **652** corresponding to a jacket accessory, accessory **654** corresponding to a pants accessory, and accessory **656** correspond to a boots accessory. Each of accessory **652-656** may be represented as its own respective accessory token. Such accessories **652-656** may be rendered on the base avatar. For example, such accessories **652-656** may be rendered on the base avatar, locally, on user device **102**.

[0124] Similar to the above, as those skilled in the art understand, such avatars **600** and **650** may be used outside of a metaverse environment. For example, a user can use their avatar **600** or **650** on other platforms, such as, but not limited to, a video game environment (e.g., a custom character in a game).

[0125] FIG. **7** is a block diagram **700** illustrating server system **104**, according to example embodiments. As shown, block diagram **700** may illustrate components of server system **104** involved in monitoring digital content for a trigger event.

[0126] Monitoring module **704** may be configured to monitor a user's digital content to determine when a trigger event **702** occurs. As provided above, a trigger event may be associated with a buying, selling, or transferring of token **512** corresponding to the digital content. In some embodiments, monitoring module **704** may be configured to monitor the user's digital content within the metaverse environment. In some embodiments, monitoring module **704** may be configured to monitor the user's digital content in one or more non-fungible token marketplaces **108**.

[0127] Responsive to identifying event **702**, monitoring module **704** may retrieve the corresponding token **512** from storage location **208**. Broadcast module **214** may then be configured to broadcast or write token **512** to blockchain **122**. In some embodiments, broadcast module **214** may directly write token **512** to blockchain **122**. For example, in some embodiments, broadcast module **214** may deploy token **512** on Ethereum Mainnet. Once deployed, token **512** may be bridged off Ethereum Mainnet to cheaper L2s. Such bridging may make it easier for users to interact directly from an L2.

[0128] Once the avatar image is minted, token **512** may be provided back to user device **102** for storage in the user's wallet **116**. In some embodiments, token **512** may be temporarily held or controlled by an entity associated with server system **104**. For example, token **512** may initially be deployed from and owned by the entity, thus allowing a development team to retain control until token **512** is transferred from a wallet associated with server system **104** (e.g., Gnosis Multisig) to a

wallet **116** of the user. Once stored in the user's wallet **116**, the user can transfer token **512** to the buyer.

[0129] Monitoring module **704** may further be configured to monitor a user's content to determine when to reward the user with additional points. As referenced above, the user's content may be displayed on the user's profile within metaverse environment **124** and/or one or more non-fungible token marketplaces **108**. Other users may interact with the user's content. Certain activities or interactions can be rewarded with a higher score, ad points, and/or metaverse tokens. Engagements such as likes, shares, views/plays, and comments can create a financial incentive for the user to create content.

[0130] In some embodiments, engagements with the user's content may be represented on the blockchain. For example, interactions with social content, such as, but not limited to, likes, shares, and views may be represented on the blockchain.

[0131] In some embodiments, interactions with social content may be represented on the blockchain by attaching a transaction to those interactions. In some embodiments, a system where interactions are ported to the blockchain automatically or manually on a preset basis may be used. In some embodiments, the interactions may be ported in a bulk transaction, with all interactions that have happened within computing environment **100** over the past period. In some embodiments, the interactions may be tokenized in real-time or near real-time. Within this structure, the gas fees associated with the transaction representations on the blockchain could be pushed to an entity associated with server system **104** or to the users. In some embodiments, there may be a threshold for how many interactions should occur for a transaction to be triggered.

[0132] In some embodiments, monitoring module **704** may update the metadata associated with the user's content to reflect updated scores. For example, as recited above, the metadata file may include various attributes associated with the token to which the metadata is linked. In some embodiments, part of the attributes may include the score for the digital content. Monitoring module **704** may update the metadata file in storage location **208** without affecting the linked token.

[0133] FIG. **8** illustrates an exemplary graphical user interface (GUI) **800**, according to example embodiments. GUI **800** may include digital content **802**. As shown, digital content **802** may include the user's avatar **804** with accessory **806**. Accessory **806** may be representative of a sword that is rendered with avatar **804**.

[0134] Via GUI **800**, other users may perform one or more actions that may provide the user with a higher score for their post. For example, a user can like or love digital content **802** via graphical element **807** or share the user's post via panel **808**. As shown in panel **808**, a user may share digital content **802** to a variety of sources that include, but are not limited to, other individuals via text message, other users within metaverse environment **124**, Facebook messenger, Facebook, WhatsApp, and the like. Additionally, in some embodiments, users may be able to save digital content **802** via graphical element **810**.

[0135] FIG. **9A** is a block diagram illustrating server system **104**, according to example embodiments. As shown, block diagram may illustrate components of server system **104** involved in indexing events occurring within metaverse environment **124**, according to example embodiments.

[0136] As shown, monitoring module **704** may be configured to monitor a user's digital content **903** in order to index interactions **901** associated with digital content **903**. Example interactions with digital content **903** may include, but are not limited to, liking, commenting, sharing, etc. the user's digital content. In some embodiments, monitoring module **704** may be configured to record each interaction **901** in association with a token **512** corresponding to the digital content. In some embodiments, recording each interaction **901** in association with a token corresponding to digital content **903** may include monitoring module **704** updating a metadata file **902** associated with token **512**. For example, monitoring module **704** may update metadata file **902** stored in storage

location **208**.

[0137] In some embodiments, monitoring module **704** may perform such indexing in real-time or near real-time on an interaction-by-interaction basis. In some embodiments, monitoring module **704** may perform batch processing of multiple interactions periodically throughout the day.

[0138] Such indexing may be useful, for example, when transferring ownership of token **512** between users. For example, assume that token **512** corresponds to a social media post that includes a thousand likes and a hundred comments. When an owner of token **512** transfers ownership of token **512** to a buyer, indexing of interactions may allow for the transfer of the interactions with token **512**. Thus, all likes, comments, or other interactions stored in association with token **512** may be transferred to the buyer.

[0139] FIG. **9B** is a block diagram illustrating server system **104**, according to example embodiments. As shown, block diagram may illustrate components of server system **104** involved in indexing events occurring within metaverse environment **124**, according to example embodiments.

[0140] As shown, monitoring module **704** may be configured to monitor a user's digital content **903** in order to index interactions **901** associated with digital content **903**. Example interactions with digital content **903** may include, but are not limited to, liking, commenting, sharing, etc. the user's digital content. In some embodiments, monitoring module **704** may be configured to record each interaction **901** in association with a token **512** corresponding to the digital content. In some embodiments, recording each interaction **901** in association with a token corresponding to digital content **903** may tokenization module **210** generating a token **950** or immutable record corresponding to each interaction **901**. Broadcast module **214** may broadcast token **950** to blockchain **122**. In some embodiments, broadcast module **214** may create a link between token **512** and token **950**.

[0141] FIG. **10A** is a block diagram illustrating interaction between a Web2 environment **1000** and a Web3 environment **1050**, according to example embodiments. As shown, monitoring module **704** may sit logically between Web2 environment **1000** and Web3 environment **1050**.

[0142] As discussed above, metaverse platform **120** may be configured to host metaverse environment **124** in which a user may generate social media posts and in which other users may interact with (e.g., comment, like, etc.) the generated social media posts. As those skilled in the art understand, a user of the metaverse environment may link their metaverse account (e.g., Web3 account) with their associated non-metaverse accounts (e.g., Web2 accounts). Example Web2 accounts may include, but are not limited to, Instagram, Twitter, TikTok, Facebook, SoundCloud, Medium, and the like.

[0143] When such linkage occurs, monitoring module **704** may be configured to monitor the user's Web2 accounts (e.g., Web2 Account **1002**), such that monitoring module **704** may port over the user's Web2 posts and/or interactions. For example, upon a user linking their Instagram account with their metaverse account, monitoring module **704** may be configured to index and port over the user's Instagram posts. Using a more specific example, assume that a user has a post (e.g., Web2 post **1004**) in Instagram. Monitoring module **704** may identify Web2 post **1004** and all of Web2 post's **1004** interactions **1006** (e.g., likes, comments, shares). Monitoring module **704** may cause metaverse platform **120** to generate a token **1040** corresponding to the user's Instagram post, as discussed above in conjunction with FIG. **7**. Similar to the process discussed above, monitoring module **704** may store a representation of the user's Instagram post in storage location **208**.

Monitoring module **704** may then generate a token **1040** based on the user's Instagram post. Such token **1040** may thus exist as a social media post in metaverse environment **124**.

[0144] In some embodiments, monitoring module **704** may further index each interaction **1006** associated with the Instagram post. For example, monitoring module **704** may be configured to identify and save each interaction **1006** in association with token **1040**. In some embodiments, such process may include monitoring module **704** updating a metadata file associated with token **1040**

(e.g., stored in storage location **208**) to include each interaction. In such embodiments, monitoring module **704** may effectively port over Web2 post's **1004** interactions **1006** from Web2 environment **1000** to Web3 environment **1050**.

[0145] In some embodiments, monitoring module **704** may be further configured to continually or periodically monitor Web2 post **1004** to continue the indexing process. For example, monitoring module **704** may continually or periodically Web2 post **1004** to update token's **1040** metadata file with new interactions. In this manner, any interaction with Web2 post **1004** may be indexed in storage location **208** such that the user may be credited with those interactions in the Web3 version of the social media post.

[0146] FIG. **10B** is a block diagram illustrating interaction between a Web2 environment **1000** and a Web3 environment **1050**, according to example embodiments. Similar to the process discussed above in conjunction with FIG. **10A**, monitoring module **704** may further be configured to monitor a user's Web3 posts, such that monitoring module **704** may index and port over the user's Web3 posts and/or interactions into Web2 environment **1000**. For example, as shown, a user may create a social media post **1082** in Web3 environment **1050**. As discussed above in conjunction with FIGS. **2**, metaverse platform **120** may generate a token (e.g., token **512**) associated with social media post **1082**. Similarly, as discussed above in conjunction with FIG. **7**, monitoring module **704** may monitor social media post **1082** to index interactions **1084** with social media post **1082**.

[0147] In some embodiments, upon a user linking their Web2 account with their Web3 account, monitoring module **704** may also be configured to port over the user's Web3 posts (e.g., social media post **1082**) into Web2 environment **1000**. Continuing with the above example, monitoring module **704** may identify social media post **1082** and interactions **1084** (e.g., likes, comments, shares). Monitoring module **704** may generate a Web2 post **1086** corresponding to social media post **1082** and may cause Web2 post **1086** to exist in Web2 environment **1000**.

[0148] In this manner, monitoring module **704** may facilitate seamless integration between social media posts existing in Web2 environment **1000** and social media posts existing in Web3 environment **1050**.

[0149] FIG. **11A** is a block diagram illustrating a computing environment **1100**, according to example embodiments. As shown, computing environment **1100** may include a Web2 environment **1100** and a Web3 environment **1150**, according to example embodiments. As shown, monitoring module **704** may sit logically between Web2 environment **1100** and Web3 environment **1150**.

[0150] Web3 environment **1150** may include a post **1102** generated by a user in metaverse environment **124**. In some embodiments, post **1102** may include a Web2 counterpart existing in Web2 environment **1100**. As discussed above in conjunction with FIG. **2**, when a user goes through the process of tokenizing a social media post, metaverse platform **120** may create a watermark to include in the social media post. As shown, post **1102** may include associated watermark **1104**. In this manner, unauthorized uses of the social media post can be tracked. Monitoring module **704** may be configured to scour or scan Web3 environment **1150** and Web2 environment **1100** to identify any posts that include the watermark unique to post **1102**. If, for example, monitoring module **704** identifies a re-post **1110** or unauthorized post that includes watermark **1104**, monitoring module **704** may attribute any interactions with the re-post or unauthorized post back to post **1102**, similar to the indexing process discussed above in conjunction with FIGS. **9A-10B**. In some embodiments, monitoring module **704** may notify owner of post **1102** of the re-post or unauthorized use.

[0151] FIG. **11B** is a block diagram illustrating a computing environment **1100**, according to example embodiments. As shown, computing environment **1100** may include a Web2 environment **1100** and a Web3 environment **1150**, according to example embodiments. As shown, monitoring module **1154** may sit logically between Web2 environment **1100** and Web3 environment **1150**.

[0152] Web3 environment **1150** may include a post **1162** generated by a user in metaverse environment **124**. In some embodiments, post **1162** may include a Web2 counterpart existing in

Web2 environment **1100**. Rather than utilize the watermarking approach discussed above in conjunction with FIG. 2 and FIG. 11A, in some embodiments, monitoring module **1154** may include an image recognition module **1156**. Image recognition module **1156** may be representative of one or more computer vision or artificial intelligence techniques for identifying reposts or unauthorized posts of post **1162**. In this manner, unauthorized uses of the social media post can be tracked using computer vision and/or artificial intelligence technology. For example, monitoring module **1154** may be configured to scour or scan Web3 environment **1150** and Web2 environment **1100** to identify any posts that may be similar to or derivatives of post **1162**. If, for example, monitoring module **1154** identifies a re-post or unauthorized post **1110** that is similar to or a derivative of post **1162**, monitoring module **1154** may attribute any interactions with the re-post or unauthorized post **1164** back to post **1162**, similar to the indexing process discussed above in conjunction with FIGS. 9A-10B.

[0153] FIG. 12 is a block diagram illustrating a computing environment **1202** interacting with server system **104**, according to example embodiments. As discussed above, a user's avatar may be used outside of metaverse environment **124**, such as in different metaverse environments or non-metaverse environments. For example, a user can use their avatar on other platforms, such as, but not limited to, a video game environment (e.g., a custom character in a game), a streaming service, and the like. For example, a user may use their avatar as a drop-in replacement for themselves in a video conference, stream, live render, etc.

[0154] When a user wishes to use their avatar in computing environment **1202** (e.g., another metaverse environment or non-metaverse environment), computing environment **1202** may identify a three-dimensional model or base avatar that is usable in or compatible with computing environment **1202**. In some embodiments, the selected three-dimensional model or base avatar may be of a file type that is compatible with computing environment **1202**. In some embodiments, server system **104** may rig the avatar in accordance with the requirements of computing environment **1202**. In some embodiments, a user may indicate certain metadata and/or accessories to follow their avatar to computing environment **1202**. For example, As discussed above, metadata **1204** may define the customizable portions of the user's avatar, i.e., those customizations that make the user's avatar unique to the user. The user may customize their avatar for use in computing environment **1202** by adjusting the customizable portions of their avatar (e.g., gender, body type, and/or skin tone) or exporting it with accessories usable in computing environment **1202**. In this manner, server system **104** may provide computing environment **1202** with the customizations, as well as the three-dimensional model or base avatar.

[0155] Using a specific example, if computing environment **1202** is Minecraft or Sandbox, server system **104** may identify a three-dimensional model or base avatar that is of a file type compatible with the Minecraft environment or Sandbox environment. In some embodiments, server system **104** may further rig the avatar in accordance with the requirements of the Minecraft environment or Sandbox environment, such that the avatar may be usable (i.e., its movements are able to be controlled by the user) and appear native to its intended environment. In this manner, the user can use their avatar or avatar accessories across different metaverse environments or non-metaverse environments.

[0156] Using another example, a user may connect their y00ts non-fungible token (NFT) to metaverse environment **124**. These NFTs are represented both on Solana, a layer-1 blockchain, and Polygon, a layer-2 blockchain. By doing so, they can showcase their unique y00t within the virtual space, allowing individuals to interact as and embody their y00t avatar regardless of which blockchain they utilize. Another example could involve a user connecting their digital artwork NFT, such as a piece from the Beeple collection, to metaverse environment **124**. Once connected, the user may display their artwork within a virtual gallery, allowing other users to appreciate the art and potentially engage in discussions about it. They may also interact or collaborate with the artist/painting in real-time. A user might own a virtual real estate NFT within a different metaverse,

such as Decentraland. They could connect their virtual property to the metaverse environment **124**, creating a bridge between the two metaverse platforms and allowing for easier navigation and interaction between them. They may view that virtual real estate as their unique identity while using out technology to view and interact with the property in real time. A user may connect their virtual fashion NFTs, like digital sneakers or apparel, to metaverse environment **124**. By doing so, their avatar within the metaverse can wear the unique digital clothing, showcasing their style and potentially attracting attention from other users. A user might have an NFT representing a famous car, such as a limited-edition virtual Lamborghini. They could connect this NFT to metaverse environment **124** and use the car as a means of transportation within the virtual space, adding another layer of interactivity and personalization. They may also be able to bring the car into a virtual space and paint it, interact, or drive it using their avatar and our motion capture. A user could connect their NFT ticket for a virtual concert or event to metaverse environment **124**. By doing so, they can attend the event within the metaverse platform and interact with other attendees, while also porting their digital identity or other NFTs into the environment, creating a more immersive and engaging experience. In some cases, a user may own an NFT representing a unique virtual pet or creature, like an Axie from Axie Infinity. They can connect this NFT to metaverse environment **124**, allowing them to interact with their pet, train it, or even engage in battles or competitions with other users' pets. By providing users with the ability to connect various types of NFTs to metaverse environment **124**, metaverse platform **120** enhances the overall user experience and encourages cross-chain or cross-metaverse interaction and collaboration.

[0157] As those skilled in the art understand, the reverse process is also possible. In some embodiments, a user could take their non-native non-fungible tokens into metaverse environment **124**. For example, a user may connect their non-fungible token to metaverse environment **124** such that their non-fungible token may be utilized within metaverse environment **124**. Using a more specific example, a user may utilize their Bored Ape non-fungible token with metaverse environment **124**. Thus, in this manner, metaverse platform **120** may provide users with cross-chain or cross-metaverse functionality.

[0158] FIG. **13** is a flow diagram illustrating a method **1300** of minting digital content, according to example embodiments. Method **1300** may begin at step **1302**.

[0159] At step **1302**, metaverse platform **120** may identify an avatar token associated with a user. In some embodiments, the avatar token may be pre-generated by metaverse platform **120** and transferred to the user. For example, metaverse platform **120** may be configured to generate a plurality of avatar tokens to assign to users of the platform. Metaverse platform **120** may transfer or assign ownership of the avatar tokens from an entity associated with server system **104** to the users.

[0160] In some embodiments, metaverse platform **120** may identify an avatar token responsive to receiving an avatar image from user device **102**. For example, as described above in conjunction with FIG. **2**, a user device **102** may upload or provide metaverse platform **120** with an avatar image indicative of an avatar of the user. Based on the avatar image and a metadata file corresponding to the avatar image, metaverse platform **120** may tokenize the avatar image provided. For example, tokenization module **210** may generate a smart contract based on the avatar image. In some embodiments, the smart contract may be representative of an ERC721 token. Metaverse platform **120** may broadcast or write the avatar token to blockchain **122**. Once broadcast or written to blockchain **122**, the avatar token may be provided to user device **102** for storage in the user's cryptographic wallet.

[0161] At step **1304**, metaverse platform may receive a request to mint digital content within metaverse environment **124**. For example, metaverse platform **120** may receive the digital content and metadata information corresponding to the digital content from user device **102**. Upon receiving the digital content, metaverse platform **120** may store the digital content at a network accessible location. Upon storing the digital content in storage location **208**, metaverse platform **120** may generate or receive a URL indicating the storage location of the digital content. Metaverse

platform **120** may provide the URL of the digital content to the user of user device **102**. The user may use the URL of the digital content to generate a metadata file corresponding to the digital content.

[0162] The digital content produced by the user may include the user's avatar. For example, a user may utilize AR software **114** to generate a video or image of the user and replace the representation of the user in the video or image with the user's avatar or insert the user's avatar into the video or image. In order to link the user's avatar token with the digital content, the metadata file may further include the URI corresponding to the tokenized avatar.

[0163] At step **1306**, metaverse platform **120** may generate a digital content token corresponding to the digital content. For example, metaverse platform **120** may generate a smart contract based on the digital content and the avatar token. In some embodiments, the smart contract may further be based on any indicated accessories in the request. In some embodiments, the smart contract may be representative of an ERC721 token. In some embodiments, the smart contract may be representative of an ERC1155 token. The smart contract may include, for example, a URI corresponding to token **512**, the URL that identifies a storage location of the digital content in storage location **208**, and the URL that identifies a storage location of the metadata file corresponding to the digital content. In some embodiments, the smart contract may further include a URI corresponding to token **212**, the URL that identifies a storage location of the avatar image in storage location **208**, and the URL that identifies a storage location of the metadata file corresponding to the avatar image. In some embodiments, the smart contract may further include a URI corresponding to token **412**, the URL that identifies a storage location of the accessory image in storage location **208**, and the URL that identifies a storage location of the metadata file corresponding to the accessory image.

[0164] At step **1308**, metaverse platform **120** may host the digital content within metaverse environment **124**. In some embodiments, in addition to hosting the digital content within metaverse environment **124**, metaverse platform **120** may further facilitate the hosting of the digital content via one or more social media applications and/or non-fungible token marketplaces **108**.

[0165] At step **1310**, metaverse platform **120** may detect an interaction with the digital content. As provided above, in some embodiments, a user's digital content token may not be written or broadcast to blockchain **122** until a user interacts (e.g., buys, sells, or transfers) with the digital content token. In other words, the tokenization referred above in step **1306** may be a “gasless” mint. Accordingly, metaverse platform **120** may monitor the digital content to determine when a trigger occurs that warrants writing the digital content token to blockchain **122**. In some embodiments, metaverse platform **120** may monitor the digital content within the metaverse environment **124**. In some embodiments, metaverse platform **120** may monitor the digital content in one or more non-fungible token marketplaces **108**.

[0166] At step **1312**, metaverse platform **120** may write the digital content token to blockchain **122** based on detecting the interaction. Responsive to identifying the interaction, metaverse platform **120** may retrieve the corresponding digital content token **512** from storage. Metaverse platform **120** may then broadcast or write the digital content token to blockchain **122**. In some embodiments, metaverse platform **120** may directly write token **512** to blockchain **122**.

[0167] FIG. **14** illustrates an example graphical user interface (GUI) **1400**, according to example embodiments. GUI **1400** may illustrate a screen capture of user device **102**, when user device **102** is utilizing full body mode of AR software **114**. In such embodiments, artificial intelligence module **115** may identify a plurality of anchor points on the user's body. In this manner, AR software **114** may be able to capture the hands and rest of the user's body more accurately.

[0168] FIG. **15** illustrates an example graphical user interface (GUI) **1500**, according to example embodiments. GUI **1500** may illustrate a screen capture of user device **102**, when user device **102** is utilizing selfie mode of AR software **114**. In such embodiments, artificial intelligence module **115** may identify a subset of anchor points from a plurality of anchor points on the user's body. For

example, because AR software **114** is in selfie mode, artificial intelligence module **115** may ignore certain anchor points, such as those associated with the user's hands and rest of their body. In this manner, AR software **114** may be able to capture the hands and rest of the user's body more accurately. In this manner, artificial intelligence module **115** may instruct AR software **114** to focus primarily on face anchor points of the user.

[0169] FIG. **18** is flow diagram illustrating a method **1800** of generating digital content, according to example embodiments. Method **1800** may begin at step **1802**.

[0170] At step **1802**, client device **102** may capture markerless motion data of a user. In some embodiments, client device **102** may utilize a camera **119** of client device **102**. In some embodiments, client device **102** may utilize one or more other sensors that may be available to client device **102**. Exemplary sensors may include, but are not limited to, Light Detection and Ranging (LIDAR) sensors, Time-of-Flight (ToF), infrared cameras, lasers, flood illuminator, proximity sensors, and dot projectors.

[0171] In some embodiments, capturing markerless motion data of the user may include client device **102** capturing multiple sets of data, depending on the distance between the user and camera **119**. For example, as discussed above in conjunction with FIG. **1**, AR software **114** may be configured such that, when user device **102** is in close range of the user (e.g., when the user's face fills up a threshold percentage of the screen), artificial intelligence module **115** may be configured to lock the chest of the user to the bottom of the screen. Once the user's body is locked in, AR software **114** may then prioritize focusing on the face and hands of the user, instead of searching for the rest of the user's body. In some embodiments, rather than locking the user's body, there may be a smooth transition when the user moves from close range to far range or vice versa.

[0172] Accordingly, when the user is within a threshold distance of camera **119** (e.g., in selfie mode), AR software **114** may capture a first portion of the markerless motion data of the user. The first portion of the markerless motion data of the user may include a first plurality of points of the user to focus on. For example, the first plurality of points may include more detailed points of the user's face.

[0173] When the user is outside of the threshold distance of camera **119** (e.g., full body mode), AR software **114** may capture a second portion of the markerless motion data of the user. The second portion of the markerless motion data of the user may include a second plurality of points of the user to focus on. For example, the second plurality of points may include points associated with the user's full body (e.g., arms, legs, hands, fingers, etc.). In some embodiments, AR software **114** may ignore or disregard a portion of the first plurality of points captured when the user is within the threshold distance of camera **119**. In other words, in selfie mode, the user may be able to open and close their mouth and eyes, smile, or frown; but in full body mode, the user may not be able to make as many facial expressions as they otherwise would in selfie mode because AR software **114** may focus less on processing data points on the user's face in order to capture the hands and rest of the user's body more accurately.

[0174] At step **1804**, client device **102** may retarget the first plurality of points and the second plurality of points to a three-dimensional model of an avatar associated with the user. As discussed above, a user may have an avatar token that uniquely identifies the user. Client device **102** may identify the user's avatar based on the user's avatar token stored in wallet **116**. Once the user's avatar is identified and/or verified, client device **102** may retarget the first plurality of points and the second plurality of points to the avatar, such that the three-dimensional model of the avatar is animated. In some embodiments, the animation may be performed in real-time or near real-time.

[0175] At step **1806**, client device **102** may render a video of the animation. In some embodiments, client device **102** may render the video of the animation local to the computing system. In some embodiments, client device **102** may cause server system **104** to remotely render the video of the animation. Such online/offline process may allow the user to view a preview of the video, while server system **104** may generate a higher quality version of the video.

[0176] At step **1808**, client device **102** may cause server system **104** to generate a non-fungible token of the video. For example, client device **102** may cause server system **104** to generate a token, as discussed above in conjunction with FIG. 5.

[0177] FIG. **16A** illustrates an architecture of system bus computing system **1600**, according to example embodiments. One or more components of system **1600** may be in electrical communication with each other using a bus **1605**. System **1600** may include a processor (e.g., one or more CPUs, GPUs or other types of processors) **1610** and a system bus **1605** that couples various system components including the system memory **1615**, such as read only memory (ROM) **1620** and random access memory (RAM) **1625**, to processor **1610**. System **1600** can include a cache of high-speed memory connected directly with, in close proximity to, or integrated as part of processor **1610**. System **1600** can copy data from memory **1615** and/or storage device **1630** to cache **1612** for quick access by processor **1610**. In this way, cache **1612** may provide a performance boost that avoids processor **1610** delays while waiting for data. These and other modules can control or be configured to control processor **1610** to perform various actions. Other system memory **1615** may be available for use as well. Memory **1615** may include multiple different types of memory with different performance characteristics. Processor **1610** may be representative of a single processor or multiple processors. Processor **1610** can include one or more of a general purpose processor or a hardware module or software module, such as service **1** **1632**, service **2** **1634**, and service **3** **1636** stored in storage device **1630**, configured to control processor **1610**, as well as a special-purpose processor where software instructions are incorporated into the actual processor design. Processor **1610** may essentially be a completely self-contained computing system, containing multiple cores or processors, a bus, memory controller, cache, etc. A multi-core processor may be symmetric or asymmetric.

[0178] To enable user interaction with the system **1600**, an input device **1645** can be any number of input mechanisms, such as a microphone for speech, a touch-sensitive screen for gesture or graphical input, keyboard, mouse, motion input, speech and so forth. An output device **1635** (e.g., a display) can also be one or more of a number of output mechanisms known to those of skill in the art. In some instances, multimodal systems can enable a user to provide multiple types of input to communicate with system **1600**. Communication interface **1640** can generally govern and manage the user input and system output. There is no restriction on operating on any particular hardware arrangement and therefore the basic features here may easily be substituted for improved hardware or firmware arrangements as they are developed.

[0179] Storage device **1630** may be a non-volatile memory and can be a hard disk or other type of computer readable media that can store data that are accessible by a computer, such as magnetic cassettes, flash memory cards, solid state memory devices, digital versatile disks, cartridges, random access memories (RAMs) **1625**, read only memory (ROM) **1620**, and hybrids thereof.

[0180] Storage device **1630** can include services **1632**, **1634**, and **1636** for controlling the processor **1610**. Other hardware or software modules are contemplated. Storage device **1630** can be connected to system bus **1605**. In one aspect, a hardware module that performs a particular function can include the software component stored in a computer-readable medium in connection with the necessary hardware components, such as processor **1610**, bus **1605**, output device **1635** (e.g., a display), and so forth, to carry out the function.

[0181] FIG. **16B** illustrates a computer system **1650** having a chipset architecture, according to example embodiments. Computer system **1650** may be an example of computer hardware, software, and firmware that can be used to implement the disclosed technology. System **1650** can include one or more processors **1655**, representative of any number of physically and/or logically distinct resources capable of executing software, firmware, and hardware configured to perform identified computations. One or more processors **1655** can communicate with a chipset **1660** that can control input to and output from one or more processors **1655**. In this example, chipset **1660** outputs information to output **1665**, such as a display, and can read and write information to storage

device **1670**, which can include magnetic media, and solid-state media, for example. Chipset **1660** can also read data from and write data to storage device **1675** (e.g., RAM). A bridge **1680** for interfacing with a variety of user interface components **1685** can be provided for interfacing with chipset **1660**. Such user interface components **1685** can include a keyboard, a microphone, touch detection and processing circuitry, a pointing device, such as a mouse, and so on. In general, inputs to system **1650** can come from any of a variety of sources, machine generated and/or human generated.

[0182] Chipset **1660** can also interface with one or more communication interfaces **1690** that can have different physical interfaces. Such communication interfaces can include interfaces for wired and wireless local area networks, for broadband wireless networks, as well as personal area networks. Some applications of the methods for generating, displaying, and using the GUI disclosed herein can include receiving ordered datasets over the physical interface or be generated by the machine itself by one or more processors **1655** analyzing data stored in storage device **1670** or **1675**. Further, the machine can receive inputs from a user through user interface components **1685** and execute appropriate functions, such as browsing functions by interpreting these inputs using one or more processors **1655**.

[0183] It can be appreciated that example systems **1600** and **1650** can have more than one processor **1610** or be part of a group or cluster of computing devices networked together to provide greater processing capability.

[0184] While the foregoing is directed to embodiments described herein, other and further embodiments may be devised without departing from the basic scope thereof. For example, aspects of the present disclosure may be implemented in hardware or software or a combination of hardware and software. One embodiment described herein may be implemented as a program product for use with a computer system. The program(s) of the program product define functions of the embodiments (including the methods described herein) and can be contained on a variety of computer-readable storage media. Illustrative computer-readable storage media include, but are not limited to: (i) non-writable storage media (e.g., read-only memory (ROM) devices within a computer, such as CD-ROM disks readably by a CD-ROM drive, flash memory, ROM chips, or any type of solid-state non-volatile memory) on which information is permanently stored; and (ii) writable storage media (e.g., floppy disks within a diskette drive or hard-disk drive or any type of solid state random-access memory) on which alterable information is stored. Such computer-readable storage media, when carrying computer-readable instructions that direct the functions of the disclosed embodiments, are embodiments of the present disclosure.

[0185] It will be appreciated to those skilled in the art that the preceding examples are exemplary and not limiting. It is intended that all permutations, enhancements, equivalents, and improvements thereto are apparent to those skilled in the art upon a reading of the specification and a study of the drawings are included within the true spirit and scope of the present disclosure. It is therefore intended that the following appended claims include all such modifications, permutations, and equivalents as fall within the true spirit and scope of these teachings.

Claims

1. A method comprising: identifying, by a computing system, images of a user performing a movement, the images comprising markerless motion capture data of the user performing the movement; extracting, by the computing system, a plurality of reference points from the markerless motion capture data of the user performing the movement, the plurality of reference points comprising references points related to a body, a face, and hands of the user performing the movement; retargeting, by the computing system, the plurality of reference points to a model of an avatar associated with the user, and wherein retargeting the plurality of reference points animates the model of the avatar to perform the movement performed by the user as the user performs the

movement; and rendering, by the computing system, a video comprising the markerless motion capture data of the user retargeted to the model of the avatar causing a body, a face, and hands of the avatar to be animated simultaneously to reflect movements of the body, the face, and the hands of the user performing the movement.

2. The method of claim 1, further comprising: generating, by the computing system, a second video having a higher quality than the rendered video.

3. The method of claim 1, wherein rendering, by the computing system, a video comprising the markerless motion capture data of the user retargeted to the model of the avatar causing the body, the face, and the hands of the avatar to be animated simultaneously to reflect movements of the body, the face, and the hands of the user performing the movement comprises: generating a preview of content in the video.

4. The method of claim 1, wherein the plurality of reference points comprises a first set of reference points captured using a first modality and a second set of reference points captured using a second modality.

5. The method of claim 4, wherein the first modality is based on the user being within a threshold distance of a camera capturing the images of the user.

6. The method of claim 5, wherein the second modality is based on the user being outside the threshold distance of the camera capturing the images of the user.

7. The method of claim 6, wherein the first set of reference points comprises at least one reference point not included in the second set of reference points and wherein the second set of reference points includes at least one reference points not included in the first set of reference points.

8. The method of claim 6, further comprising: merging, by the computing system, the first set of reference points and the second set of reference points.

9. A non-transitory computer readable medium comprising one or more sequences of instructions, which, when executed by a processor, causes a computing system to perform operations comprising: identifying, by the computing system, images of a user performing a movement, the images comprising markerless motion capture data of the user performing the movement; extracting, by the computing system, a plurality of reference points from the markerless motion capture data of the user performing the movement, the plurality of reference points comprising references points related to a body, a face, and hands of the user performing the movement; retargeting, by the computing system, the plurality of reference points to a model of an avatar associated with the user, and wherein retargeting the plurality of reference points animates the model of the avatar to perform the movement performed by the user as the user performs the movement; and rendering, by the computing system, a video comprising the markerless motion capture data of the user retargeted to the model of the avatar causing a body, a face, and hands of the avatar to be animated simultaneously to reflect movements of the body, the face, and the hands of the user performing the movement.

10. The non-transitory computer readable medium of claim 9, further comprising: generating, by the computing system, a second video having a higher quality than the rendered video.

11. The non-transitory computer readable medium of claim 9, wherein rendering, by the computing system, a video comprising the markerless motion capture data of the user retargeted to the model of the avatar causing the body, the face, and the hands of the avatar to be animated simultaneously to reflect movements of the body, the face, and the hands of the user performing the movement comprises: generating a preview of content in the video.

12. The non-transitory computer readable medium of claim 9, wherein the plurality of reference points comprises a first set of reference points captured using a first modality and a second set of reference points captured using a second modality.

13. The non-transitory computer readable medium of claim 12, wherein the first modality is based on the user being within a threshold distance of a camera capturing the images of the user.

14. The non-transitory computer readable medium of claim 13, wherein the second modality is

based on the user being outside the threshold distance of the camera capturing the images of the user.

15. The non-transitory computer readable medium of claim 14, wherein the first set of reference points comprises at least one reference point not included in the second set of reference points and wherein the second set of reference points includes at least one reference points not included in the first set of reference points.

16. The non-transitory computer readable medium of claim 14, further comprising: merging, by the computing system, the first set of reference points and the second set of reference points.

17. A system comprising: a processor; and a memory having programming instructions stored thereon, which, when executed by the processor, causes the system to perform operations comprising: identifying images of a user performing a movement, the images comprising markerless motion capture data of the user performing the movement; extracting a plurality of reference points from the markerless motion capture data of the user performing the movement, the plurality of reference points comprising references points related to a body, a face, and hands of the user performing the movement; retargeting the plurality of reference points to a model of an avatar associated with the user, and wherein retargeting the plurality of reference points animates the model of the avatar to perform the movement performed by the user as the user performs the movement; and rendering a video comprising the markerless motion capture data of the user retargeted to the model of the avatar causing a body, a face, and hands of the avatar to be animated simultaneously to reflect movements of the body, the face, and the hands of the user performing the movement.

18. The system of claim 17, wherein the operations further comprise: generating a second video having a higher quality than the rendered video rendered.

19. The system of claim 17, wherein rendering the video comprising the markerless motion capture data of the user retargeted to the model of the avatar causing the body, the face, and the hands of the avatar to be animated simultaneously to reflect movements of the body, the face, and the hands of the user performing the movement comprises: generating a preview of content in the video.

20. The system of claim 17, wherein the plurality of reference points comprises a first set of reference points captured using a first modality and a second set of reference points captured using a second modality.
